

Copilot Studio Training – Day 10

Agenda:

- Get Started with MCP Server
- MCP Server Terminologies
- OOTB Available MCP Server
- Connect Third Party MCP Server
- Some Real-World Use Cases
- Multi Agent Capabilities in Copilot Studio

Model Context Protocol (MCP)

What is MCP?

- MCP stands for **Model Context Protocol**.
- Introduced by **Anthropic**.
- It is a **standardized way for LLMs to interact with tools, APIs, and knowledge bases** like databases or documents.
- MCP is like the **USB-C of AI applications** — a **universal connector** between LLMs and external resources.

Why Do We Need MCP?

- Earlier AI applications:
 - Used LLMs only (like ChatGPT) → limited to training data.
- Next step:
 - Agentic frameworks (like Copilot Studio, Azure AI Foundry) → LLMs + tools + knowledge.
- Problem:
 - Too much "glue code" (manual integration logic).
 - Hard to maintain and scale across multiple apps.

MCP solves this with a universal protocol and SDKs that make it easy to connect any tool/knowledge to LLMs.

Real-World Example

Use Case:

Generate a report comparing **Nvidia vs Tesla**:

- Description
- Financial metrics

- Summary
- Latest news

Challenge:

- LLM can summarize and describe, but **can't access live stock prices**.
- Needs **external tool/API (like Yahoo Finance)**.

Solution:

- Use MCP to integrate tools like:
 - Yahoo Finance API (as an MCP server)
 - Google Search
 - Internal databases or PDFs

MCP Components

Component	Role
LLM	Heart of the AI app; makes smart decisions
MCP Client	Acts as the interface for the LLM to discover and use tools
MCP Server	Hosts tools/resources/prompts and exposes them using a standard protocol
Tools	API functions, e.g., "search_places", "fetch_news"
Resources	Static knowledge – PDFs, files, databases
Prompts	Predefined prompts that can be reused by LLMs

Technical Workflow

1. MCP Client Startup

- Reads available MCP servers from config
- Sends list_tools, list_resources, list_prompts calls

2. MCP Server Response

- Returns descriptions + input schemas (e.g., lat/long for a map tool)
- Standardized format → LLM understands how to use it

3. LLM Executes

- Based on prompt and tool description, LLM:
 - Picks the right tool
 - Extracts parameters from natural language

- Makes a call
- Summarizes response

Implementation Guide

1. Create MCP Server

- Use SDKs (e.g. .NET) to build your MCP server based on workflow needs.

2. Publish via Connector

- Define a **custom connector** using an OpenAPI YAML schema.

3. Consume Tools in Copilot Studio

- Go to “Add Tool” > “Model Context Protocol” and select your connector.
- Enable **Generative Orchestration** to use MCP tools.

Practical Use Cases

- Fetching real-time sales data from ERP systems.
- Integrating HR candidate management systems.
- Automating approval flows using MCP-connected logic modules.
- Querying knowledge graphs directly in Copilot Studio agents.

Notes & Best Practices

- MCP currently supports **Tools and Resources**. Prompts are not yet available in Copilot Studio.
- **Server-Sent Events (SSE)** support is deprecated as of August 2025 — recommended transport is **Streamable**.
- Always maintain secure endpoints and schema validation to ensure robust integrations.

Here’s a structured and simplified set of class notes based on the video about **Model Context Protocol (MCP)**:

Example: Google Maps + Todoist Bot

- LLM receives: “I’m going hiking in Lonavala, show me places and create todo”
- MCP client sends tool descriptions of:
 - Google Maps (search places)
 - Todoist (create task)
- LLM figures out:
 - Tool to call: search_places

- Extracted params: location = Lonavala
- Follow-up tool: create_task on Todoist

Key Takeaways

- MCP simplifies tool/knowledge integration with LLMs
- Reduces repetitive glue code
- Enables scalable, modular AI application design
- Early-stage tech with high future potential

Case Study 01 – Connect your first Custom MCP Server within Copilot Studio

Let's connect readymade Microsoft Learn MCP Server for asking some questions in the natural language:

Microsoft Learn MCP Server Endpoint URL: <https://learn.microsoft.com/api/mcp>

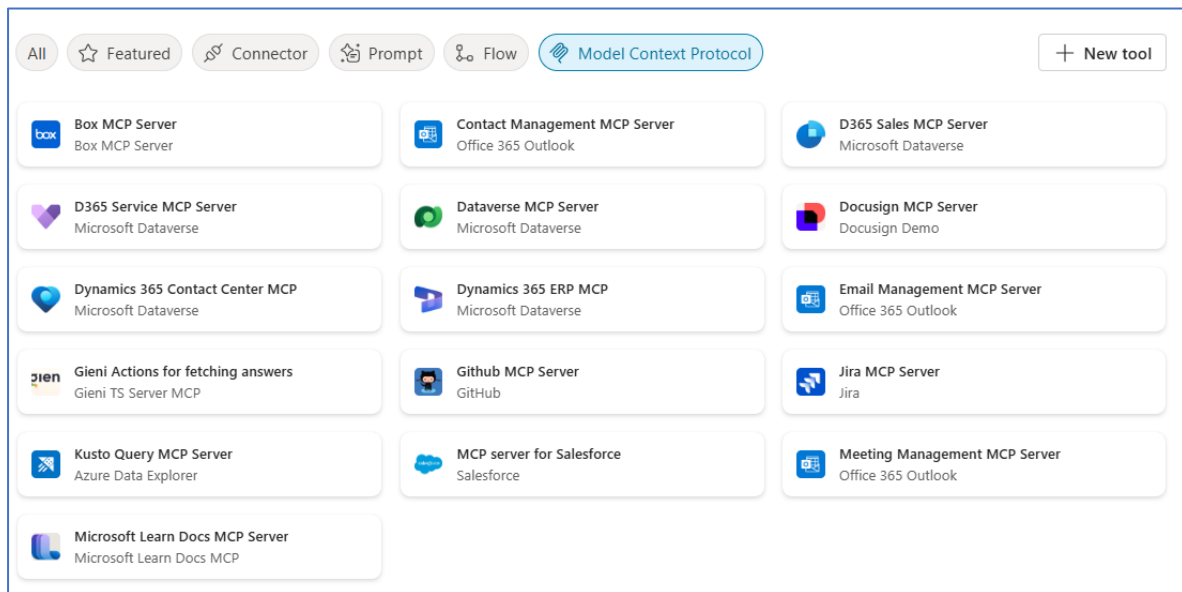
Reference:

<https://learn.microsoft.com/en-us/training/support/mcp>

<https://learn.microsoft.com/en-us/microsoft-copilot-studio/agent-extend-action-mcp>

Case Study 02 – Explore Native MCP Servers available in the Copilot Studio

Copilot Studio natively offer different MCP Servers.



Case Study 03: Smart Calendar Assistant with MCP in Copilot Studio

Objective

Build or enhance an existing **Copilot Studio Agent** to function as a **Smart Calendar Assistant** using **Meeting Management MCP Server**, enabling it to:

- View calendar meetings
- Check calendar availability
- Schedule meetings
- Cancel meetings

⚙️ How to Set It Up

1. Create or Use Existing Copilot Studio Agent

- Go to your [Copilot Studio](#)
- Choose or create a **new agent**

2. Add MCP Tool

- Navigate to: **Tools > Add a Tool**
- Select: **Model Context Protocol** tab
- Choose: **Meeting Management MCP Server**
- Provide: Existing or new connection

3. Available MCP Servers by Microsoft

MCP Server	Use Case
Contact Management MCP	Access and manage Outlook contacts
Email Management MCP	Access Outlook inbox/sent items
Meeting Management MCP	View/create meetings
Dataverse MCP	Read/write to Microsoft Dataverse
Dynamics 365 MCP	ERP, CRM, Sales, Contact Center

Internals of MCP Server (Technical)

- Once the Meeting Management MCP Server is added:
 - No code needed — tools auto-configure
 - Tools like:
 - getMeetings
 - getCalendarView
 - getCalendars
 - getRoom
 - createMeeting
- Each tool is linked to a **Graph API endpoint**
- Example Tool → createMeeting: Calls POST /me/events with parameters

Sample Use Cases (Prompt + Response)

Prompt	Behind-the-Scenes Action
“Get my meetings for today”	Calls getMeetings tool (Graph API: /me/events?filter=date)
“Schedule a meeting with Contact on 17 th August between 4 to 5 PM IST”	Calls createMeeting tool, resolves user ID for Patel, creates event in calendar

 The agent understands natural language and converts:

- **Date/time** → UTC
- **User references** → Tenant users
- **Intent** → Appropriate tool/API

Computer Use Agent in Copilot Studio

1. What It Is

- **Computer use** is a preview tool in **Microsoft Copilot Studio** that enables your agent to interact with and automate tasks on a **Windows computer** (websites and desktop apps) using a **virtual mouse and keyboard**.
- It lets Copilot agents click buttons, navigate menus, type text, and interact with UI elements just like a human would — **even if no API is available**.
- Powered by **Computer-Using Agents (CUA)**, an AI model with screen perception and UI understanding.

2. Key Capabilities

- Automates tasks on **web and desktop applications**.
 - Uses **AI vision + reasoning** to adapt when UI changes (e.g., layout changes on websites).
 - No coding required — users describe their task in **natural language**.
 - Can be used for things like data entry, form filling, invoice processing, data extraction, research tasks, etc.
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3. How It Works

The tool loop involves three main phases:

1. **Perception:** Captures screenshots to understand the current screen state.
2. **Reasoning:** Evaluates goals and previous steps to decide what to do next.
3. **Action:** Performs clicks, typing, scrolling, etc., until the task completes.

It *emulates human interaction* with the UI instead of calling APIs.

4. Adding Computer Use to an Agent

From the Copilot Studio UI:

1. Add a **new tool** to your agent.
2. Select **Computer use**.
3. Provide **natural language instructions** describing the automation.
4. Configure **machine target, credentials, and other settings**.
5. Test and refine instructions.

5. Execution and Configuration

- You select a **machine** where computer use runs (local or hosted).
- Instructions can include dynamic inputs (like URLs or form fields).
- Credentials for websites or apps can be stored securely (internal store or Azure Key Vault).
- Access control can limit what the agent interacts with

6. Security & Human Supervision

- You can enable **human supervision** — reviewers verify potentially risky actions before execution continues.
- Admin controls allow organizations to manage security and compliance of computer use features

7. Limitations

- Still in **preview**; not meant for production.
- Performance varies: better on web tasks (~80% success) than desktop apps (~35% success).
- Inconsistent performance possible due to UI changes or timing variations.)
- Some UI controls (custom widgets, date pickers) can be difficult for automation.

8. Best Practice Tips

- Use **clear, detailed instructions**; include exact URLs and app names
- Restrict tasks to trusted sites and apps for safety.
- Use dedicated machines with limited permissions.

9. Billing

- While in preview, *computer use* runs consume **Copilot Credits** based on action steps performed.

Reference:

<https://learn.microsoft.com/en-us/microsoft-copilot-studio/computer-use>

Case Study: Automating Inventory Data Entry Using Computer Use in Copilot Studio

Overview

This case study demonstrates how **Computer Use** in Microsoft Copilot Studio automates manual data entry into a web-based inventory system.

The automation simulates a human interacting with a browser — navigating to a website, filling out forms, and submitting inventory records — without using APIs.

Business Scenario

Department: Operations / Inventory Management

Problem: Manual entry of new inventory items into a web-based system is time-consuming and prone to errors.

Goal: Use Computer Use to automatically submit multiple product entries into the inventory system.

Web URL:

<https://computerusedemos.blob.core.windows.net/web/Adventure/index.html>

This web application contains:

- Product Name field
- Product Code field
- Quantity field
- Price field
- Supplier field
- Submit button

Product Name	Code	Quantity	Price	Supplier
Rear Derailleur	RD-4821	50	42.75	Tailspin Toys
Pedal Set	PD-1738	80	19.99	Northwind Traders
Brake Lever	BL-2975	35	14.50	Trey Research
Chainring Bolt Set	CB-6640	100	5.25	VanArsdel, Ltd.
Bottom Bracket	BB-9320	60	24.90	Tailwind Traders

Solution Architecture

Tool Used

Computer Use tool inside Microsoft Copilot Studio

How It Works

1. Agent opens browser
2. Navigates to provided URL
3. Identifies input fields using AI screen perception
4. Types product details
5. Clicks Submit
6. Repeats process for each product

This uses:

- Screenshot-based UI understanding
- Mouse & keyboard simulation

- Step-by-step reasoning loop

Prompt:

1. Go to <https://computerusedemos.blob.core.windows.net/web/Adventure/index.html>.

2. Submit a new entry for each of the following items:

Rear Derailleur, RD-4821, 50, 42.75, Tailspin Toys

Pedal Set, PD-1738, 80, 19.99, Northwind Traders

Brake Lever, BL-2975, 35, 14.50, Trey Research

Chainring Bolt Set, CB-6640, 100, 5.25, VanArsdel, Ltd.

Bottom Bracket, BB-9320, 60, 24.90, Tailwind Traders

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